

PV-CTL control board

Rev A1 | 2026-10-05 | Schematics authored by Chinmoy Bhuyan | Cost-first PV-P75/100/110 controller: F280039C on BUS-, one trip latch, one SELV barrier

01 PC connector, supply, reset

2x32 PC contract to the power board (all LIVE): only +3V3 is used (200 mA allocated by PV-PWR); TPS3828 reset + 1.6 s watchdog

File: sheets/01_supply.kicad_sch

02 F280039C, clock, debug, EEPROM

TMS320F280039C (100-pin PZ), decoupling, 25 MHz SPXO, cTI-20 JTAG (LIVE), EEPROM, boot straps, command pull-downs, run LED

File: sheets/02_mcu.kicad_sch

03 Reference and analog front ends

REF3030E 3.0 V = VREFHI = VDAC = NTC bias = trip thresholds; IL1-4 gain stages; power-board signals into the ADC through 100 R / 1 nF; 8 NTC biases + inlet NTC

File: sheets/03_afe.kicad_sch

04 Hardware trip comparators

7 x TLV9024 on +3V3 (open drain, no firmware): IL windows from each sensor's Uref; VA / VB over-voltage, NTC over-temperature and open probe from VREF ladders

File: sheets/04_trip.kicad_sch

05 Trip latch, watchdog, gating

74LVC07 wired-AND of every trip source -> PRE_N -> 74LVC1G74 latch (no firmware); heartbeat monoflop; 74LVC08 gating of 16 PWM, EN, K_A, K_B, K_PRE, STATUS, FAN

File: sheets/05_latch.kicad_sch

06 Barrier B1: isolators, CAN, RS-485

Reinforced barrier B1: CA-IS3050W CAN, CA-IS3082WNX RS-485, 2 x CA-IS3821LG, CA-IS3820LG; bus TVS, jumper terminations, pluggable field terminals (SELV)

File: sheets/06_iso.kicad_sch

07 SELV supply, fans, enable / status

SELV 24 V input (AUX-T1 reinforced winding, 12-36 V), TPS54360B 5 V and fan supply, 3 fan headers + tach, ENABLE / STATUS terminal, SELV 0 V to PE

File: sheets/07_selv.kicad_sch

ARCHITECTURE-COSTFIRST sections 6.1 (trip sources -> one latch), 10 (two zones); PC levels as built on PV-PWR (hardware/PV-PWR/outputs/PV-PWR_design_check.txt).
Assembly option: the two TLV9024 marked 'PV-P100/110 only' are not fitted on PV-P75 (bom/PV-CTL-P75_BOM.csv).
Pin plan: gen/data/pv_ctrl_pin_plan.csv (from SPRSP61C, checked every build).

Not bench-validated. All bands and budgets calculated in design_check().
Sheet 1
Schematics authored by Chinmoy Bhuyan
LIVE zone = BUS- (hazardous live); SELV zone behind reinforced barrier B1

Sheet: /
File: PV-CTL.kicad_sch

Title: **PV-CTL control board - Cover / Hierarchy**

Size: A3

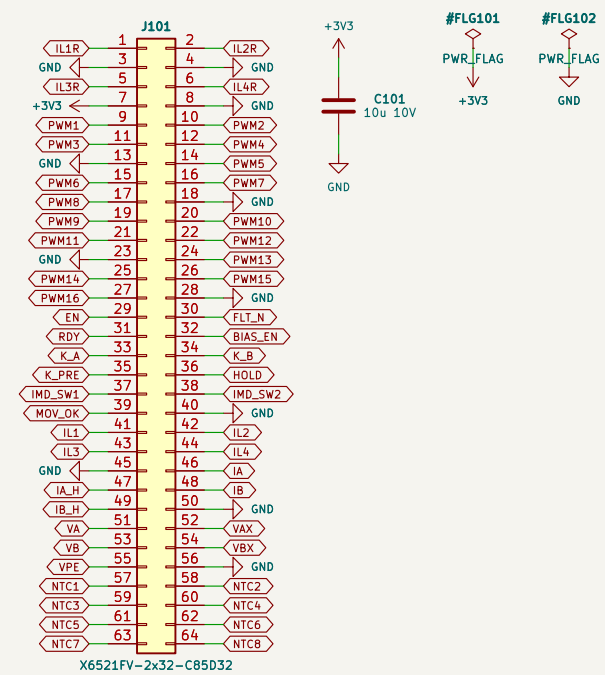
Date: 2026-10-05

Rev: **A1**

KiCad E.D.A. 10.0.6

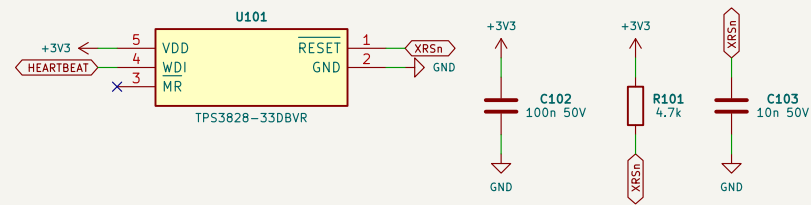
Id: 1/8

PC CONNECTOR (GEN/INTERFACES.PY PC; AGND = GND ON THIS BOARD)



LIVE: GND = AGND = BUS-; nothing on this connector leaves the enclosure. Only +3V3 comes across (the former +24V / +5V pins carry the sensor references IL1R-IL4R since PV-PWR rev A2). +3V3 feeds the MCU, so the power board's 3.3 V signals and VDDIO share one rail and one power-up order (no back-powering).

RESET AND WATCHDOG (TPS3828-33)



RESET (open drain) holds XRSn below 2.93 V and for 200 ms after; no WDI edge for 0.9-2.5 s -> reset. XRSn 4.7k pull-up and 10 nF (SPRSP61C table 5-1: 2.2-10k, <= 100 nF). XRSn also clears the heartbeat monoflop, so every MCU reset sets the trip latch. +5V / +24V supervision: on the power board (RDY).

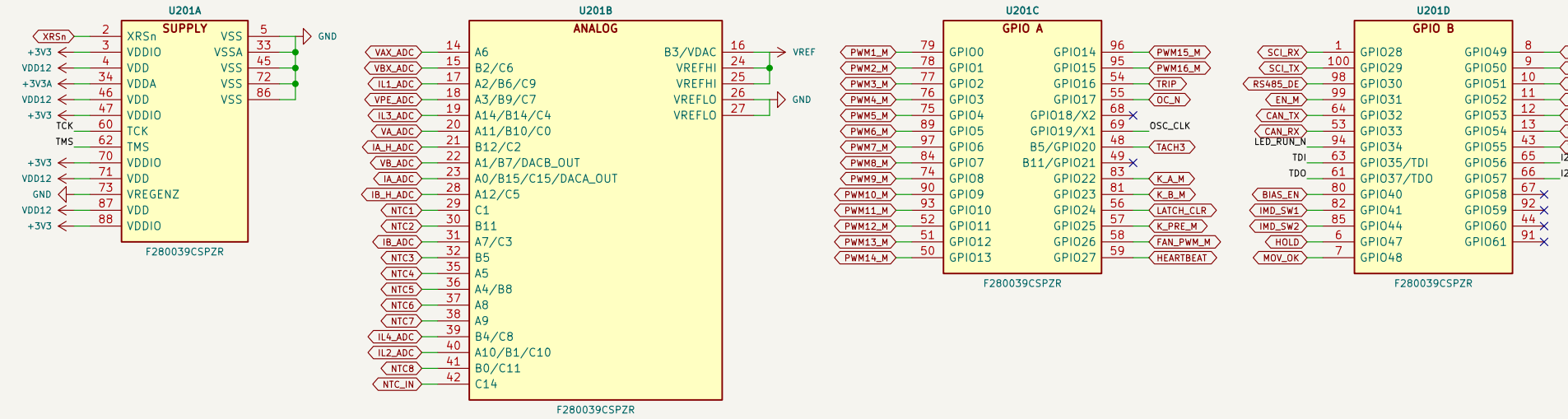
Not bench-validated. All bands and budgets calculated in design_check().
Sheet 2
Schematics authored by Chinmoy Bhuyan
LIVE zone = BUS- (hazardous live); SELV zone behind reinforced barrier B1

Sheet: /01 PC connector, supply, reset/
File: 01_supply.kicad_sch

Title: PV-CTL control board - PC connector, supply, reset

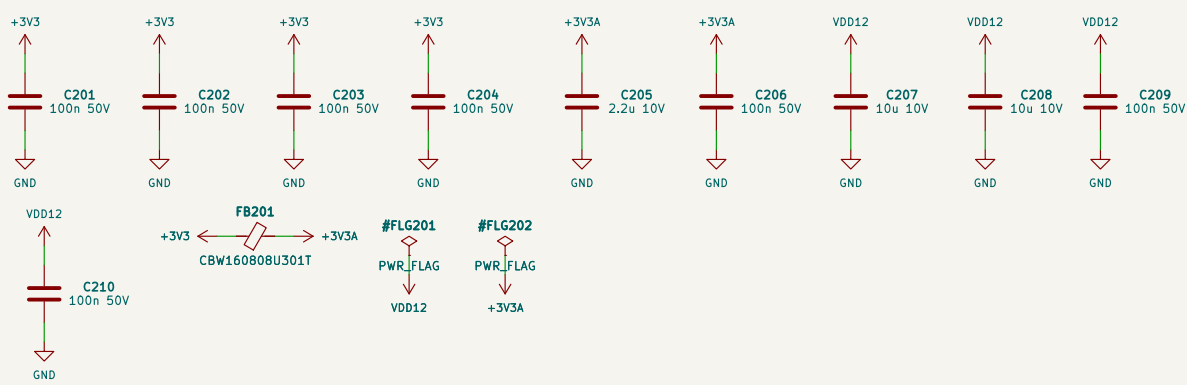
Size: A3	Date: 2026-10-05	Rev: A1
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F280039CSPZR



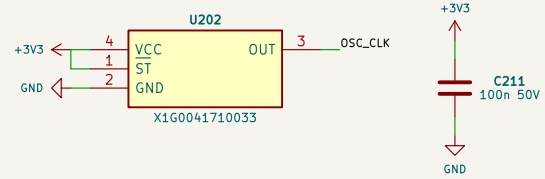
Pins as gen/data/pv_ctrl_pin_plan.csv (checked against SPRSP61C table 5-1 on every build). VREGENZ = GND; internal 1.2 V VREG; VDD only decoupled.

DECOUPLING (SPRSP61C 6.12.1.5.1)



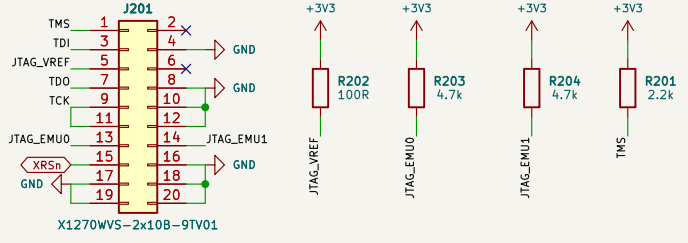
VDDIO 4 x 100 nF (>= 0.1 uF per pin); VDDA 2.2 uF + 100 nF behind a 300 R bead (pi filter allowed by 6.12.1.A.1); VDD 2 x 10 uF + 2 x 100 nF = 20.2 uF (10-26.6 uF total, internal VREG, 10 % parts).

25 MHZ CLOCK (X1 SINGLE-ENDED)



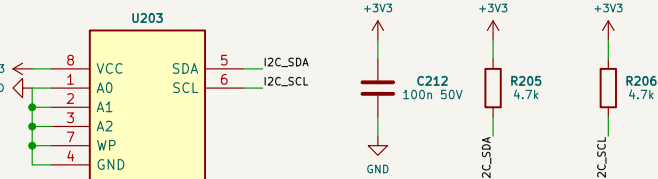
SG-2105TF +/-50 ppm (CAN bit timing; CAN boot needs the external clock). f(X1) 10-25 MHz, tr/td <= 6 ns (SPRSP61C 6.12.3.2.1).

JTAG CTI-20 - HAZARDOUS LIVE (ISOLATED PROBE ONLY)



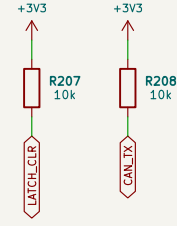
Label: 'HAZARDOUS VOLTAGE - up to 1000 V DC to earth - Isolated debug probe only'. TMS 2.2k pull-up (no TRSTn on F28003x, table 5-1); TRST (2) open; EMU0/1 4.7k pull-up; RESET (15) on XRSn; 17-19 GND. Field firmware update over CAN (DCAN boot option 1 or application loader); no live access needed.

EEPROM (CALIBRATION, SERIAL NUMBER, EVENT LOG)



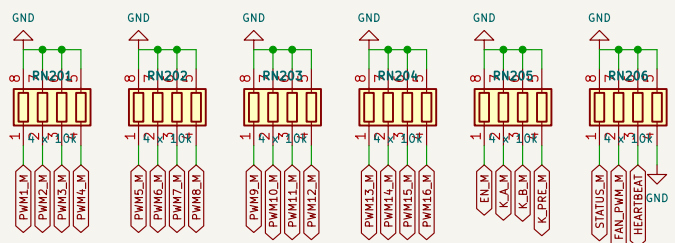
BL24C256A at 0x50 (A2-A0 low). WP low (writable; firmware guards the calibration pages). I2C 4.7k pull-ups.

BOOT STRAPS



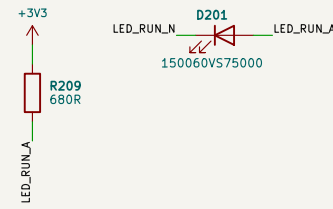
Flash boot = GPIO24 high and GPIO32 high at reset (SPRSP61C table 7-8); 10k pull-ups on LATCH_CLR (GPIO24) and CAN_TX (GPIO32; also recessive while the MCU is in reset).

COMMAND PULL-DOWNS (6 X 4 X 10K ARRAYS)



Every MCU output that ends in an AND gate here is low while the MCU is in reset or unprogrammed (GPIOs high impedance); 16 PWM_EN, K_A, K_B, K_PRE, STATUS, FAN_PWM, HEARTBEAT (23 lines, 1 spare element). BIAS_EN, IMD_Sw1/2 go straight to PC; pulled down on the power board (10k / 100k, PV-PWR).

RUN LED



Green. GPIO34 low = on (firmware heartbeat). 680 R; ~2 mA.

Not bench-validated. All bands and budgets calculated in design_check().
Sheet 3
Schematics authored by Chinmoy Bhuyan
LIVE zone = BUS- (hazardous live); SELV zone behind reinforced barrier B1

Sheet: /02 F280039C, clock, debug, EEPROM/
File: 02_mcu.kicad_sch

Title: PV-CTL control board - F280039C, clock, debug, EEPROM

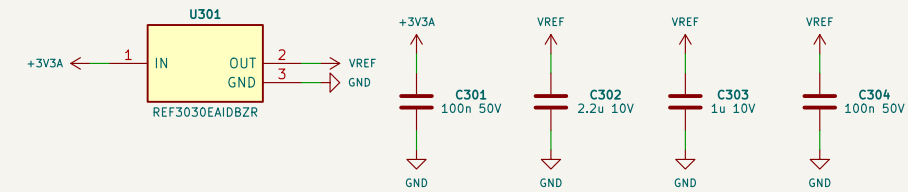
Size: A2 Date: 2026-10-05

Rev: A1

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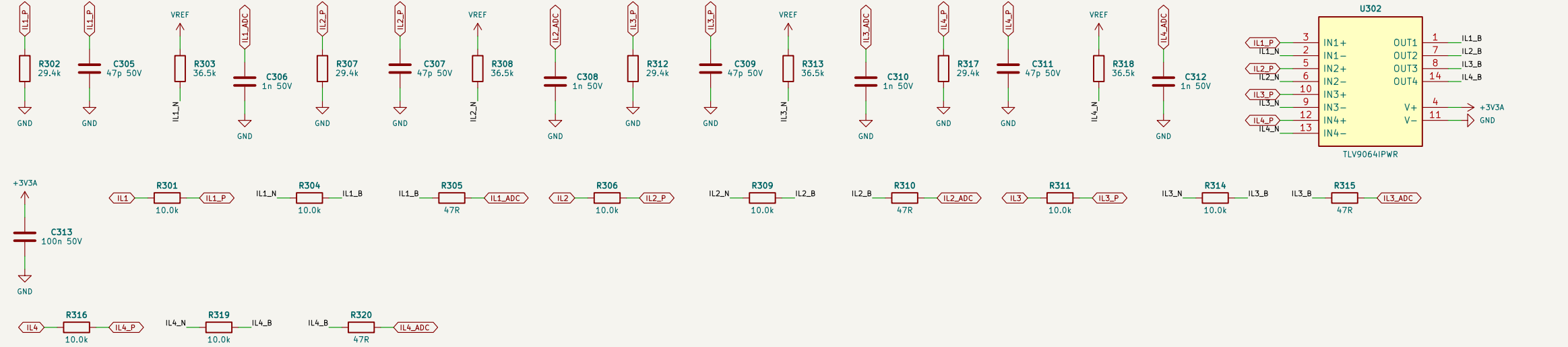
Id: 3/8

3.0 V REFERENCE (REF3030E FROM +3V3A)



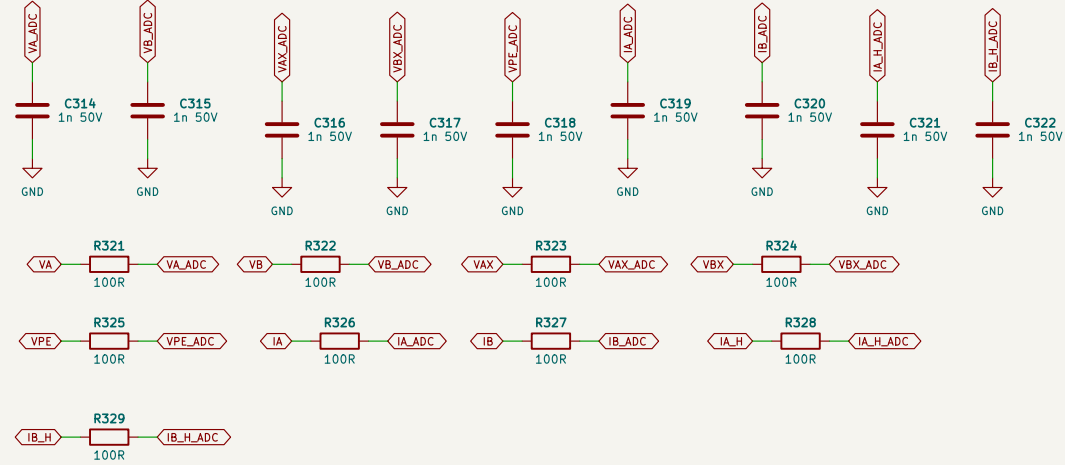
Feeds VREFHI (pins 24/25, >= 2.2 uF). VDAC (pin 16, >= 1 uF), the NTC biases, the IL offset and the threshold ladders of sheet 04. ADC, CMPSS and hardware trips are ratiometric to one reference. It runs from VDDA, so it never sits above VDDA (SPRSP61C 6.12.1.4.2). Needs +3V3 >= 3.20 V (REF30E dropout).

INDUCTOR-CURRENT FRONT ENDS IL1-IL4 (TLV9064 GAIN STAGES)



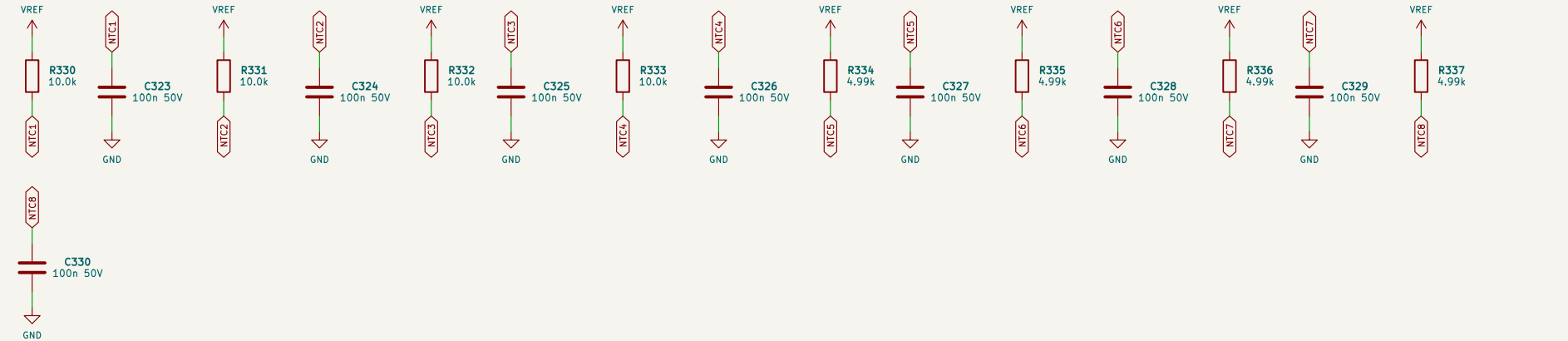
ILn = 2.5 V + 10.667 mV/A (STK-HO/A 75 on PV-PWR, 0.5-4.5 V, not-ratiometric) -> 10.0k / 29.4k 0.1 %
-> ILn_P (47 pF, a = 0.740); trip window (sheet 04) and the + Input: Rg 36.5k to VREF; Rf 10.0k; ADC = 1.555 V + 10.14 mV/A (0.072 A/LSB, -153...+142 A) -> 47 R / 1 nF -> ADC + CMPSS; IL4 = 0 V on PV-P75.

POWER-BOARD ANALOG SIGNALS (VMID 1.641-1.651 V PER PORT, NOT ON PC)



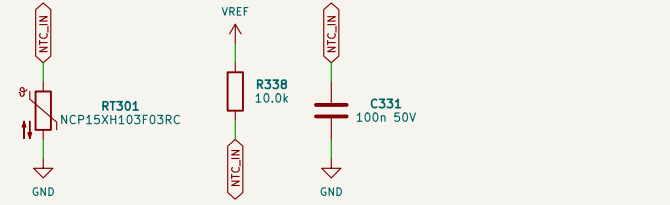
VA/VB/VAX/VBX/VPE = VMID + V/1203; IA/IB = VMID + 3.0 mV/A (+/-478 A); IA_H/IB_H = VMID + 1.0 mV/A. Anti-alias poles are on the power board: here 100 R / 1 nF CGO as the ADC charge bucket. The trip comparators take the PC net itself (no ADC kick-back on them).

NTC INPUTS (10 K B25/100 3988 K TO AGND + 100 NF ON PV-PWR, BIASED HERE FROM VREF)



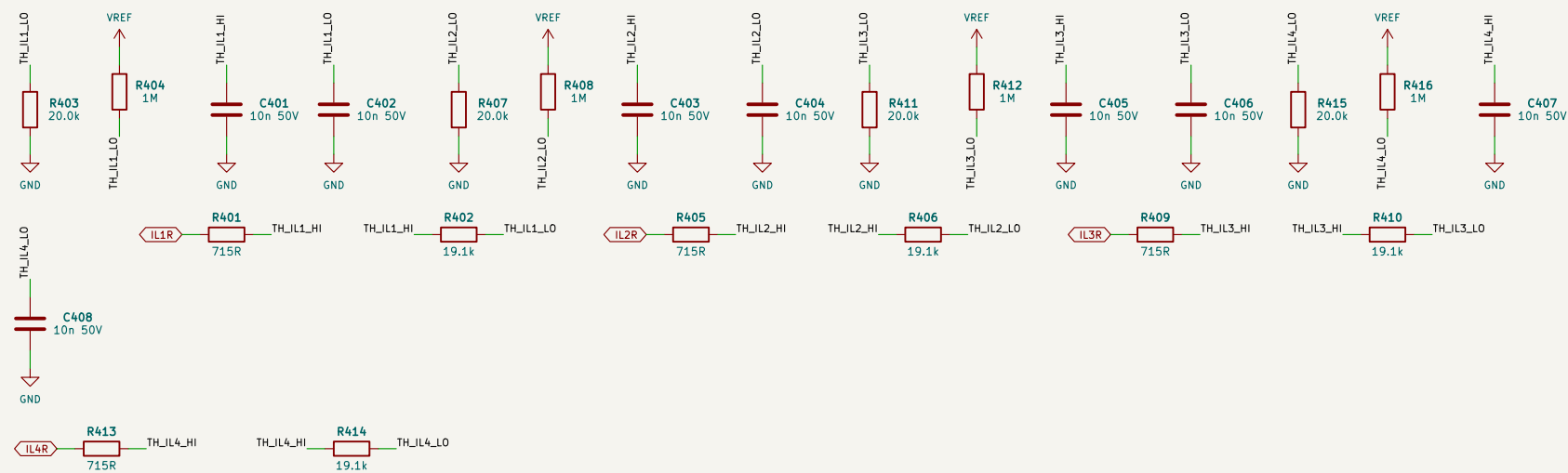
NTC1-4 heatsink sections (10.0k). NTC5-8 Inductor windings (4.99k), 100 nF each. Open probe = 3.0 V; trips through the open-probe comparators (sheet 04); short = 0 V; trips as over-temperature.

INLET-AIR NTC (ON BOARD)



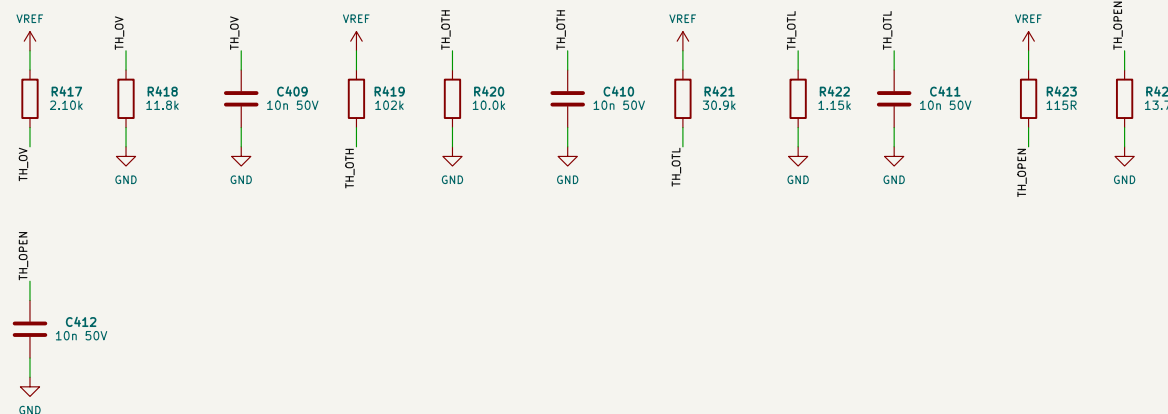
Murata NCP15XH103F03RC, 10.0k from VREF; ADC only (fan law, derating).

IL THRESHOLD LADDERS (0.1 %, 25 PPM/K). ONE PER PHASE FROM THAT SENSOR'S UREF (ILNR ON PC)



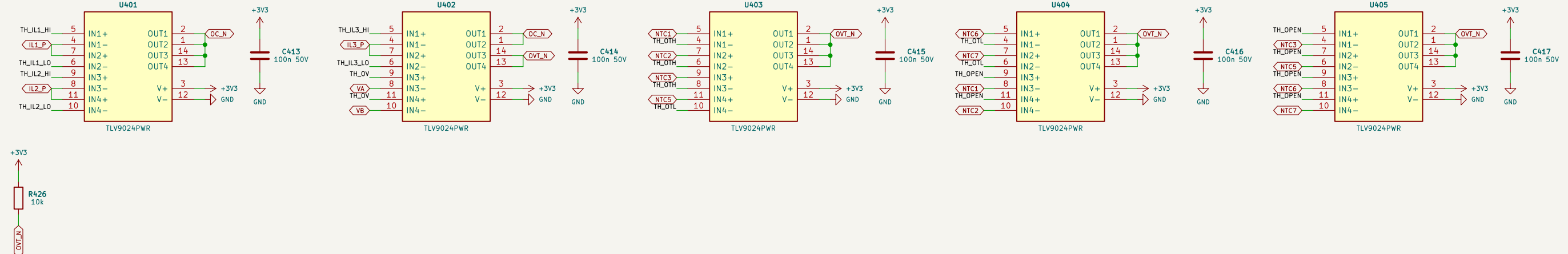
ILnR = 715R - TH_ILn_HI = 19.1k - TH_ILn_LO = 20.0k - GND; the window follows the sensor's own zero (Vref 2.48-2.52 V is no longer an error term; Vref +/- 0 mV is), 1 M from VREF to TH_ILn_LO, with the sensor dead (IL = ILR = 0 V) TH_ILn_LO stays -30 mV above ILn_P and the phase trips. Phase 4: fitted on every build.

THRESHOLD LADDERS (0.1 %, 25 PPM/K, FROM VREF)



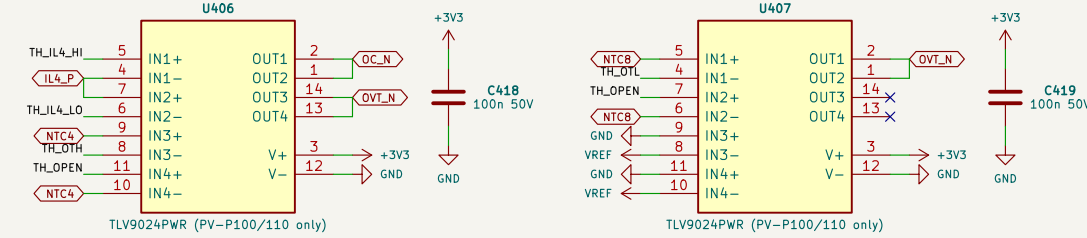
TH_LOV: VA/VB 1084 V; TH_OTH: heatsink 87.9 C (PV-PWR: <= 89.9 C); TH_OTL: Inductor 150 C; TH_OPEN 2.975 V: an NTC node above R is an open probe (cold limit -30 C).

PHASES 1-3 AND THE PORT VOLTAGES (FITTED ON EVERY BUILD)



OC_N (wired-OR, 10k to +3V3): ILn_P above TH_ILn_HI or below TH_ILn_LO, n = 1-3. OVT_N (10k): VA, VB above TH_LOV; NTC1-3 (heatsink) below TH_OTH; NTC5-7 (inductor) below TH_OTL; any of them above TH_OPEN. Port over-current (387-413 A) arrives on FLT_N from the power board's own comparators (PV-PWR).

PHASE 4 - ASSEMBLY OPTION: FITTED ON PV-P100/110 ONLY



IL4 window, NTC4 / NTCB over-temperature and open probe. Not fitted on PV-P75 (bom/PV-CTL-P75_BOM.csv), where IL4 = 0 V and NTC4 / NTCB are open; nothing then pulls OC_N / OVT_N for phase 4. Fitted on a 4-phase build, a dead IL4 sensor (0 V) or an open NTC4 / NTCB trips (fail-safe). Firmware cross-checks.

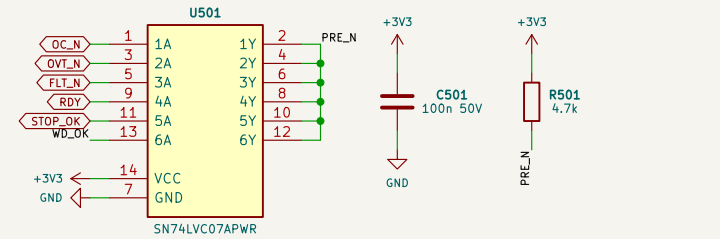
Not bench-validated. All bands and budgets calculated in design_check().
Sheet 5
Schematics authored by Chinmoy Bhuyan
LIVE zone = BUS- (hazardous live); SELV zone behind reinforced barrier B1

Sheet: /04 Hardware trip comparators/
File: 04_trip.kicad_sch

Title: PV-CTL control board - Hardware trip comparators

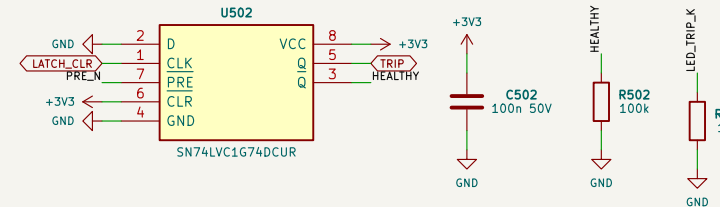
Size: A2 Date: 2026-10-05 Rev: A1
KiCad E.D.A. 10.0.6 Id: 5/8

TRIP SOURCES --> PRE_N (OPEN-DRAIN WIRED-AND, 4.7K TO +3V3)



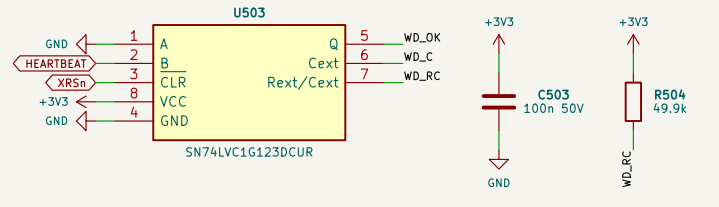
74LVC07 (loff, 5 V tolerant): DC_N, OVT_N, FLT_N (driver DESAT/UVLO + port OC of PV-PWR), RBY (gate supplies, +5V, live 24 V on PV-PWR), STOP_OK (ENABLE), WD_OK (heartbeat and MCU reset). Any low -> PRE_N low -> latch set asynchronously (Q = TRIP = 1).

THE LATCH (74LVC1G74, SET-DOMINANT)



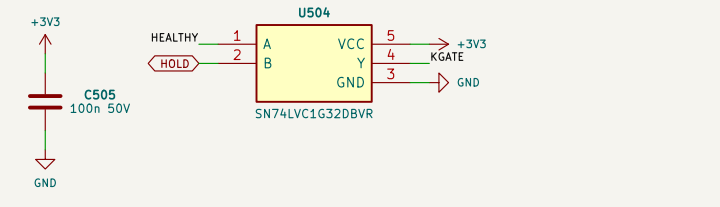
PRE = PRE_N, CLR tied high, D = GND, CLK = LATCH_CLR (GPIO24). A firmware rising edge loads 0 (clears) only while PRE_N is high (no source active); with any source active PRE wins (Table 9-1). Q = TRIP (red LED, INPUT/SAFETY -> T23), Q = HEALTHY (gate). Unpowered: loff -> HEALTHY reads 0 (100%).

HEARTBEAT WATCHDOG (74LVC1G123)



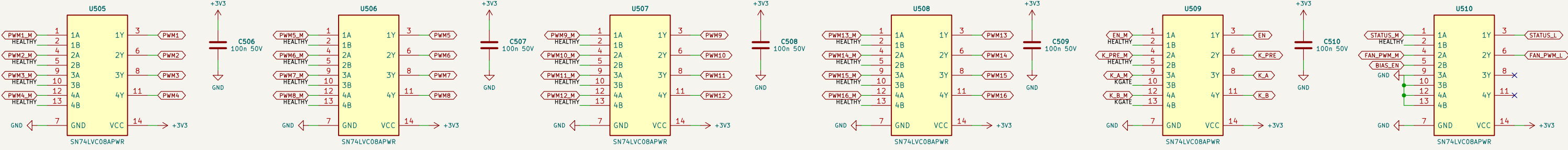
B = HEARTBEAT (rising edges from the control ISR, >= 500 Hz). A = GND, CLR = XRSn, Q = WD_OK stays high while edges come faster than tw = 49.9k x 100n (4.7-5.8 ms); no edge, a stuck pin or any MCU reset (XRSn low) -> WD_OK low -> latch. Rext to VCC, Cext between Cext and Rext/Cext (SCE586E table 4-1).

CONTACTOR GATE: KGATE = HEALTHY OR HOLD (74LVC1G32)



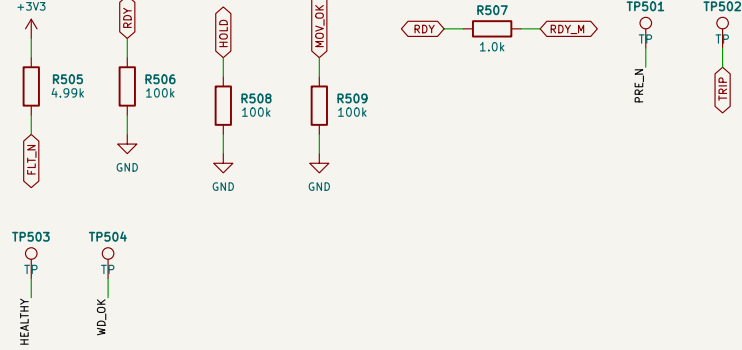
A trip drops K_PRE at once, but K_A / K_B only while HOLD is low; while the power board holds a contactor (port current above its breaking limit) this board never commands it open; the power board's coil interlock enforces the same. Closing still needs the MCU command AND (HEALTHY OR HOLD).

GATING: 6 X 74LVC08 (22 GATES USED, 2 SPARE WITH INPUTS GROUNDED)



PWMn = PWMn_M AND HEALTHY (16). EN = EN_M AND HEALTHY, K_PRE = K_PRE_M AND HEALTHY, K_A/K_B = K_n_M AND KGATE, STATUS_L = STATUS_M AND HEALTHY, FAN_PWM_L = FAN_PWM_M AND BIAS_EN (PV-PWR rule: fans only while the gate bias loads the aux supply). The power board pulls every PC input low (10k / 100k).

POWER-BOARD STATUS LINES (PV-PWR AS BUILT)



FLT_N: open drain on PV-PWR, pulled up HERE (4.99k, PV-PWR recommendation), RBY: open drain pulled up on PV-PWR (no pull-up here, 100k pull-down allowed), to the MCU via 1k, HOLD: push-pull, MCU_OK: open drain with 10k on PV-PWR; 100k pull-downs define an unplugged cable (not held / monitor open).

Not bench-validated. All bands and budgets calculated in design_check().
Sheet 6
Schematics authored by Chinmoy Bhuyan
LIVE zone = BUS- (hazardous live); SELV zone behind reinforced barrier B1

Sheet: /05 Trip latch, watchdog, gating/
File: 05_latch.kicad_sch

Title: PV-CTL control board – Trip latch, watchdog, gating

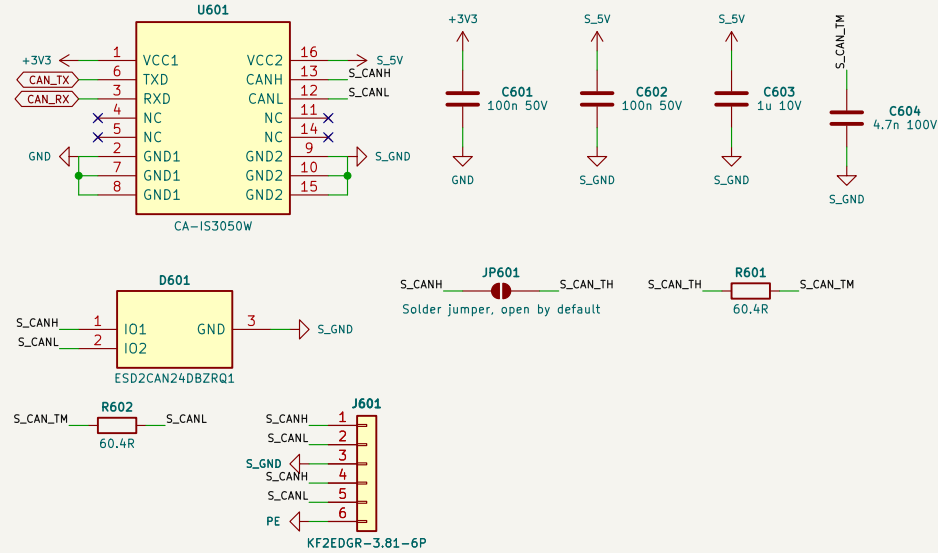
Size: A2 Date: 2026-10-05 Rev: A1

KiCad E.D.A. 10.0.6 Id: 6/8

BARRIER B1 RULES (INSULATION_SPEC 'REINFORCED HV-PELV, DC POLE')

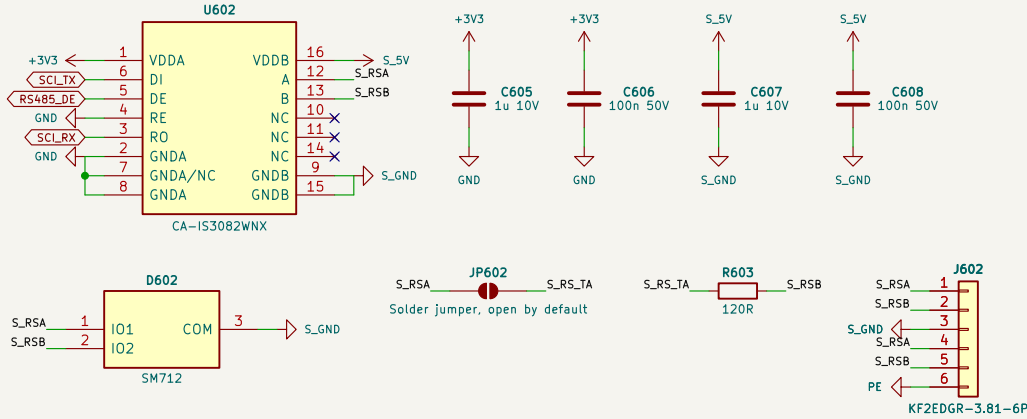
Every part on this sheet that touches LIVE and SELV is a declared isolator: VIOSM >= 8 kV, VISO >= 5 kVrms, VIOWM >= 1414 V DC, CLR/CPG 8 mm, 8 mm < 10 mm (PD2 creepage) -> conformal coating PD1 over the B1 zone is MANDATORY; routed slot under every isolator; SELV copper >= 10 mm from LIVE copper elsewhere.

ISOLATED CAN (BMS / PCS), CA-IS3050W



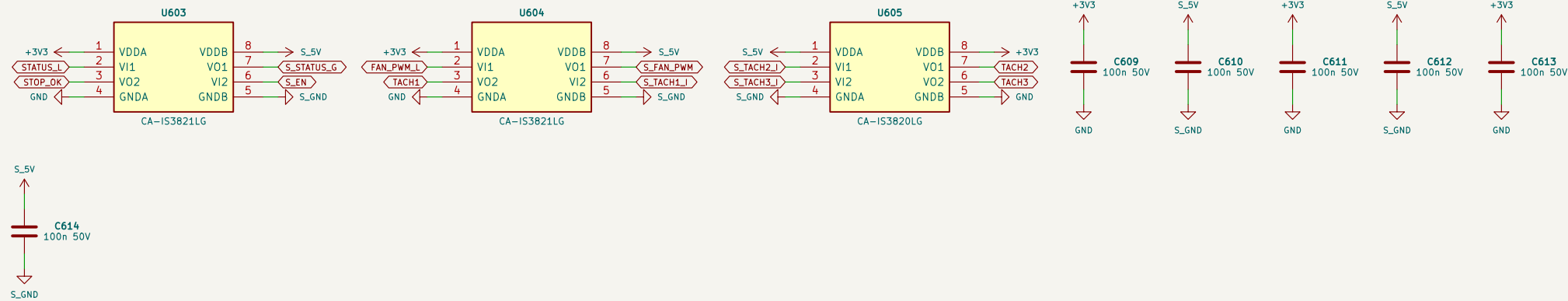
Logic side +3V3 (ICC1 <= 3.6 mA), bus side S_5V (ICC2 <= 73 mA dominant); no live +5V needed, Split termination 2 x 60.4 R + 4.7 nF by JP; ESD2CAN24 to SELV 0 V. Terminal: CANH, CANL, 0 V, CANH, CANL (in / out), shield to PE.

ISOLATED RS-485, CA-IS3082WNX



Logic side +3V3; bus side S_5V. DE from the MCU (internal pull-down; driver off in reset). RE tied low (receiver always on, fail-safe). 120 R by JP; SM712 to SELV 0 V. Terminal: A, B, 0 V twice, shield.

DIGITAL ISOLATORS (DEFAULT LOW = SAFE)



3821 #1: STATUS_L -> S_STATUS_G (out), S_EN -> STOP_OK (in; SELV side unpowered or open = STOP).
3821 #2: FAN_PWM_L -> S_FAN_PWM (out; unpowered = fans off), S_TACH1_L -> TACH1. 3820 (A side SELV):
S_TACH2_L, S_TACH3_L -> TACH2, TACH3, 100 nF per side.

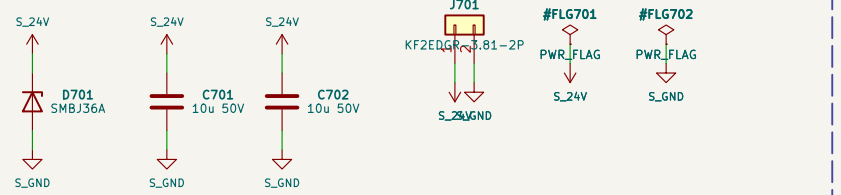
Not bench-validated. All bands and budgets calculated in design_check().
Sheet 7
Schematics authored by Chinmoy Bhuyan
LIVE zone = BUS- (hazardous live); SELV zone behind reinforced barrier B1

Sheet: /06 Barrier B1: isolators, CAN, RS-485/
File: 06_Iso.kicad_sch

Title: PV-CTL control board - Barrier B1: isolators, CAN, RS-485

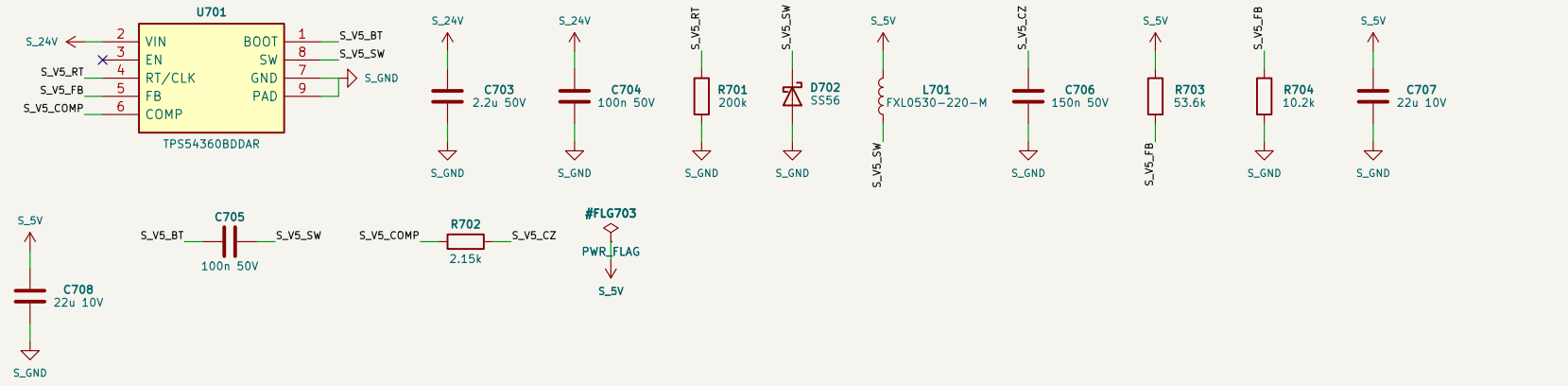
Size: A3 Date: 2026-10-05 Rev: A1
KiCad E.D.A. 10.0.6 Id: 7/8

SELV INPUT (2-PIN, DOUBLE-INSULATED LEADS, >= 10 MM FROM LIVE COPPER)



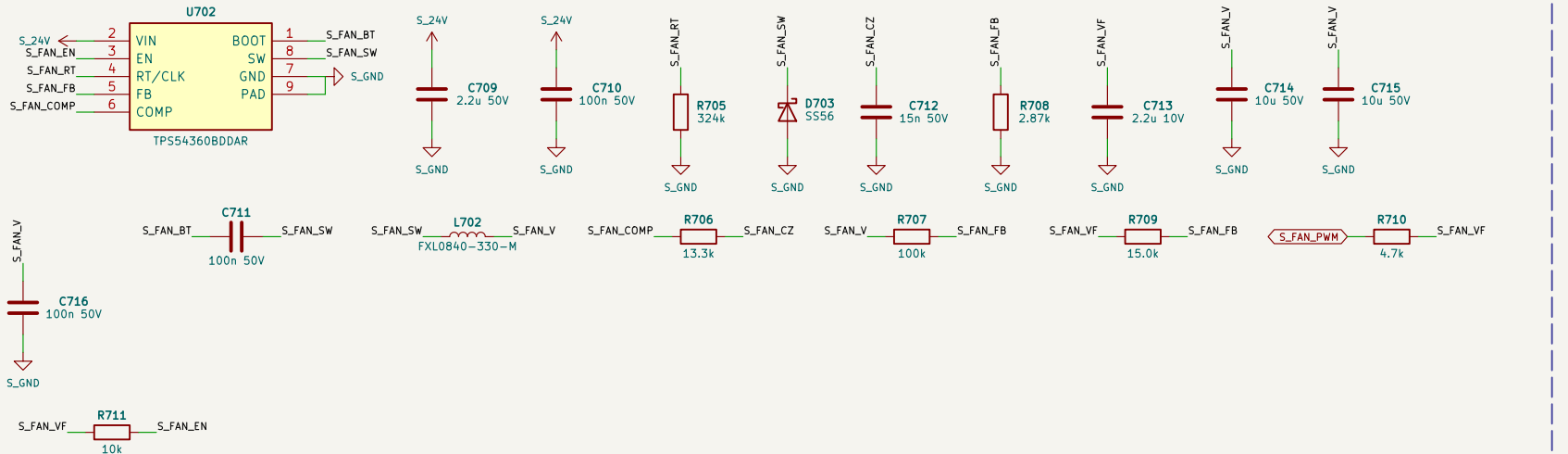
AUX-T1 fan winding, cross-regulated: 12-36 V (sags to -15 V without gate-bias load, PV-PWR), SMBJ36A clamp (36 V working), 2 x 10 uF 50 V. Both bucks have no under-voltage lock-out above 4.5 V.

SELV 5 V (TPS54360B, 500 KHZ)



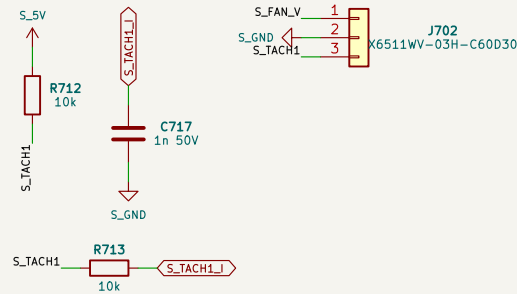
CAN and RS-485 bus sides, SELV side of the three digital isolators: EN floating = on (internal pull-up), 22 uH, 2 x 22 uF.

FAN SUPPLY (TPS54360B, 300 KHZ): SPEED BY VOLTAGE, PASSES WHAT THE WINDING GIVES



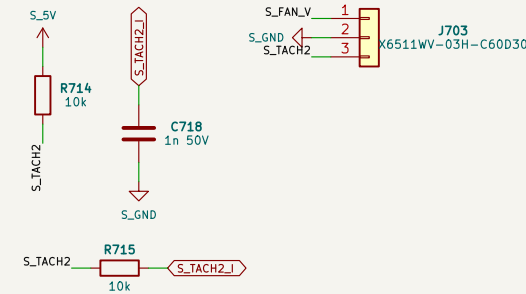
S_FAN_PWM (isolated, 0/5 V) -> 4.7k / 2.2 uF -> S_FAN_VF: EN via 10k (off below 1.1-1.3 V; duty below -25 %; static low or a dead live side = fans OFF) and current injection into FB through 15.0k; higher duty = lower voltage (24 V ... 7 V); at a low input it runs in dropout. Rtop 100k / Rbot 2.87k.

FAN 1



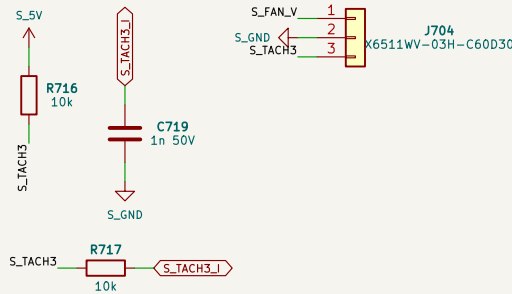
1 +V, 2 0 V, 3 tach: 10k to S_5V, 10k + 1 nF (10 us) to the Isolator Input.

FAN 2



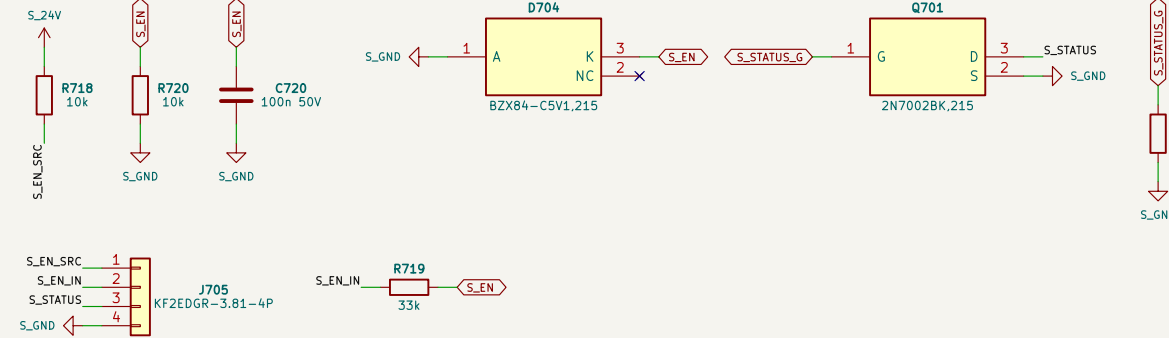
1 +V, 2 0 V, 3 tach: 10k to S_5V, 10k + 1 nF (10 us) to the Isolator Input.

FAN 3



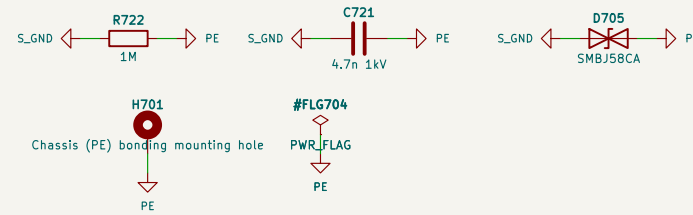
1 +V, 2 0 V, 3 tach: 10k to S_5V, 10k + 1 nF (10 us) to the Isolator Input.

ENABLE (EXTERNAL STOP) AND STATUS TERMINAL



1 S_EN_SRC (S_24V via 10k; dry-contact supply), 2 S_EN_IN (contact return or a PLC output), 3 STATUS (open drain 2N7002BK; conducting = healthy), 4 0 V, S_EN_IN -> 33k / 10k (S_L V clamp, 100 nF: 0.77 ms) -> Isolator: open contact, broken wire or lost SELV = STOP_OK low = uHch set (no firmwares).

SELV 0 V TO PE (SELV MAY BE EARTHED BY THE INSTALLER: PELV)



1 M || 4.7 nF 1 kV plus SMBJ58CA: a common-mode surge on the field cables is clamped to PE: a solid PELV bond by the installer is harmless. Plated MS hole = PE / shield bond near the terminals.

Not bench-validated. All bands and budgets calculated in design_check().
Sheet 8
Schematics authored by Chinmoy Bhuyan
LIVE zone = BUS- (hazardous live); SELV zone behind reinforced barrier B1

Sheet: /07 SELV supply, fans, enable / status/
File: 07_selv.kicad_sch
Title: PV-CTL control board – SELV supply, fans, enable / status
Size: A2 Date: 2026-10-05 Rev: A1
KiCad E.D.A. 10.0.6 Id: 8/8